

NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Software Engineering

Mid-Term Exam
(MT-173)

Max. Marks: 20

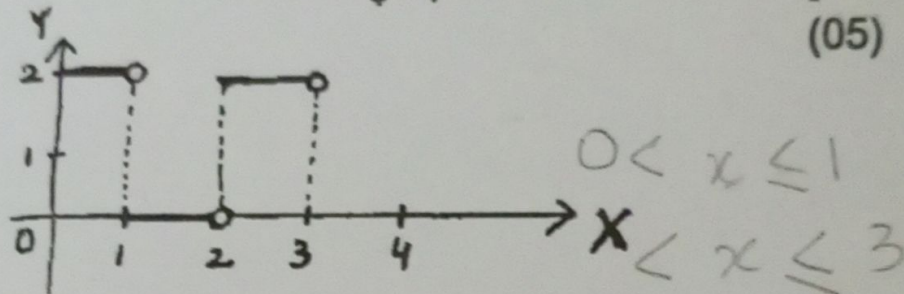
Time: 1.5 hr

Date: _____

Q. 1 Identify the formula for the function graphed.

[CLO-01]

(05)



Q. 2 Apply relevant concepts to evaluate the following indeterminate forms:

(a) $\lim_{x \rightarrow 0} \frac{\ln(\cos x)}{\ln(\cos 3x)}$

(b) $\lim_{x \rightarrow 0} x \tan\left(\frac{\pi}{2} - x\right)$

[CLO-01]

(05)

Q. 3 Let $u = \tan^{-1}\left(\frac{y}{x}\right)$ then show that u satisfies the laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

[CLO-02]

(05)

Q. 4 Assuming the possibility of expansion, find the series in $x + 1$ of:

[CLO-03]

(05)

e

$$f(x) = e^{-x}$$

1 $-e$

2 e

3 $-e$

4 e

$$e - e(x-1) + \frac{e(x-1)^2}{2} - \frac{e(x-1)^3}{6} + \frac{e(x-1)^4}{24} + \dots$$